

Laboratory Assignment: Measuring Network Performance with iperf3

Learning Objectives

- Understand and measure network performance metrics: latency, bandwidth, and throughput.
- Use iperf3 to simulate and analyze TCP and UDP traffic.
- Compare the performance across different protocols and configurations.

Background Concepts

Latency: Time taken for a packet to travel from source to destination.

Bandwidth: Theoretical maximum capacity of a network link (in Mbps or Gbps).

Throughput: Actual rate at which data is successfully transferred (often lower than bandwidth).

Lab Requirements

- Two machines (VMs, physical systems, or containers) connected over a network.
- iperf3 installed on both machines.

Setting up the Lab Environment on Windows

1. Download iperf3 for Windows from: <https://iperf.fr/iperf-download.php>
2. Extract the zip file and place the iperf3 executable in a known folder.
3. Add the folder path to the Windows environment PATH variable (optional).
4. Open Command Prompt to run iperf3 commands.
5. Ensure Windows Defender Firewall allows iperf3 traffic on port 5201.

Setting up the Lab Environment on Linux

Run the following commands:

```
sudo apt update && sudo apt install iperf3 -y
```

Step-by-Step Instructions

1. Start the iperf3 server on one machine:
`iperf3 -s`
2. On the client machine, measure TCP throughput:
`iperf3 -c <server_ip_address>`

3. Measure UDP bandwidth:

```
iperf3 -c <server_ip_address> -u -b 100M
```

4. Measure latency using ping:

```
ping <server_ip_address>
```

5. Test with varying parameters:

```
iperf3 -c <server_ip_address> -t 30
```

```
iperf3 -c <server_ip_address> -P 4
```

Sample Results Table

Metric	Protocol	Value	Tool Used
Latency	ICMP	~2 ms avg	ping
Bandwidth	UDP	~90 Mbps	iperf3
Throughput	TCP	~87 Mbps	iperf3

Lab Report Questions

1. What is the difference between bandwidth and throughput?
2. How does TCP throughput change with multiple parallel streams?
3. How did packet loss behave during the UDP test?
4. What network conditions could affect your measurements?
5. What would be the bottleneck in a real-world network scenario based on your results?

Cleanup

- Stop the iperf3 server using Ctrl+C.
- Restore firewall or antivirus rules if modified.